L-C-770B
October 11, 1974
SUPERSEDING
Fed. Spec. L-C-770A
February 3, 1967

FEDERAL SPECIFICATION

CUPS, PLASTIC; LIDS, PLASTIC DISPOSABLE CUP

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers disposable plastic cups and plastic lids for use with these cups.

1.2 Classification. The disposable plastic cups and lids shall be of the following items, types, and sizes, as specified (see 6.2). Cup sizes shown are in ounces.

Item 1 - Cup.

Type I - Standard (table I).

Size 7.

Type II - Extra insulated (tables II or IIa).

Size 6.
Size 8.
Size 10.

Item 2 - Lid.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-665 - Boxes, Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Denver, San Francisco, Los Angeles, and Seattle, WA.

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

21 CFR 121:

Subpart F - Food Additives Resulting From Contact With Containers or Equipment and Food Additives Otherwise Affecting Food.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

E 171 - Standard Atmospheres for Conditioning and Testing Materials.

(Application for copies should be addressed to the American Society for Testing and Materials 1916 Race Street, Philadelphia, PA. 19103.)

National Motor Freight Traffic Association Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The cups and lids shall be the current standard product of the manufacturer, except for any modification specified herein.

3.2 Materials. The materials used for the manufacture of cups and lids shall conform to the requirements specified herein. Materials shall be of the commercial quality for the purpose intended.

3.2.1 Compliance. The supplier shall present either a certificate or satisfactory evidence that the materials used in the finished product comply with the requirements of 3.2.2.

3.2.2 Plastic. Plastic materials used for cups and lids shall be clean, sanitary, and free of toxic ingredients and shall conform to subpart F of 21 CFR 121.

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3.2.3 Cups.

3.2.3.1 Standard. The materials for item 1, type I standard cups shall be impact polystyrene or foamed sheet polystyrene.

3.2.3.2 Extra insulated. The materials for item 1, type II extra insulated cups shall be plastic foam molded polystyrene (expanded bead type) or plastic foam molded polystyrene (expanded bead type) preformed into sheets.

3.2.4 Lids. The materials for item 2 lids shall be impact polystyrene.

3.3 Design and construction.

3.3.1 Cups. The cups shall be round, designed for nesting, with a rolled or rounded top drinking edge. The cups shall have a stacking feature to prevent severe wedging. The cups may have horizontal circumferential or vertical stiffening ribs.

3.3.1.1 Standard cups. Item 1, type I standard cups shall be one-ply, two-ply with air space, or two-ply laminated and shall be as specified in table I.

3.3.1.2 Extra insulated cups. Item 1, type II extra insulated cups shall be molded or formed from preformed plastic sheets and shall be as specified in table II or table IIa, as applicable.

3.3.2 Lids. Item 2 lids shall be of the coverall construction with a skirt or flange which extends over the brim of the cup and may be extended inside the cup. At the option of the manufacturer and in accordance with commercial practice, a vent hole may be provided in lids for relief of pressure buildup from hot liquids in cups. Lids shall be 0.008 inch minimum thickness.

TABLE I. Item 1, type I, standard cups

Nominal size	Brimful capacity	Sidewall thickness
	Min.	Min.
Fluid oz.	Fluid oz.	Inch
7	6.75	0.008

TABLE II. Item 1, type II, extra insulated cups (molded construction)

Nominal size	Brimful capacity	Sidewall thickness	Bottom thickness
	Minimum	Minimum	Minimum
Fluid oz.	Fluid oz.	Inch	Inch
6	5.75	0.040	0.075
8	7.75	0.040	0.075
10	9.75	0.040	0.075

TABLE IIa. Item 1, type II, extra insulated cups (formed from preformed sheets)

Nominal size	Brimful capacity	Sidewall thickness	Bottom thickness
	Minimum	Minimum	Minimum
Fluid oz.	Fluid oz.	Inch	Inch
6	6.5	0.025	0.040
8	9.0	0.025	0.040
10	11.0	0.025	0.040

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3.4 Insulation, type II cups. The maximum allowable outside temperature for item 1, type II extra insulated cups shall be 155°F. when tested in accordance with 4.4.2.

3.5 Rigidity, type II cups. Item 1, type II extra insulated cups shall have a maximum allowable deflection of 0.280 inches when tested in accordance with 4.4.3.

3.6 Staining, cups.

3.6.1 Brewed coffee. Cups shall have only a slight sidewall stain when tested in accordance with 4.4.4.1. Slight stain is defined here as one which is readily perceptible without close examination but is insufficient to markedly alter the original color of the sidewall.

3.6.2 Cola carbonated beverage. Cups shall have no sidewall stain when tested in accordance with 4.4.4.2.

3.7 Penetration, cups. Cups shall have no penetration of liquids to the outside wall nor shall there be any evidence of leakage when tested in accordance with 4.4.5.

3.8 Distortion, cups. Cups shall not distort when filled with brewed coffee at 180°F. ± 2°F. and shall not be susceptible to distortion from hand pressure due to normal lifting or handling of cups (see 4.4.6).

3.9 Odor, cups and lids. Cups and lids at the time of unpacking from boxes or bags shall be free from all unpleasant and objectionable odors (see 4.4.7).

3.10 Capacity, cups (item 1). Type I cups shall meet the brimful capacity requirement of table I. Type II cups shall meet the brimful capacity requirement of table II or table IIa, as applicable. Determination of conformance shall be as specified in 4.4.8.

3.11 Marking for identification. Cups and lids shall be marked in accordance with the manufacturer's commercial practice.

3.12 Workmanship. Cups and lids shall be clean, well made, and free from any defects which may affect appearance or serviceability. Lids shall fit cups snugly. Cups shall be free from warpage, cracks, chips, tears, holes, cuts, blisters, stains, discolorations, and flash.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where a certificate of compliance is submitted (see 3.2.1), the Government reserves the right to check-test such items to determine the validity of the certification.

4.2 Sampling. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.2.1 Lot. Unless otherwise specified (see 6.2), a lot shall consist of all cups and lids of one item, type, and size submitted for inspection at the same time. The sample unit shall be one completely fabricated end item.

4.2.2 Sampling for examination. Sampling for visual and dimensional examination shall be in accordance with level S-3.

4.2.3 Sampling for tests. Sampling for tests shall be in accordance with level S-3. Samples shall be selected at random from at least five different cartons from each lot.

4.2.4 Acceptable quality levels (AQL). AQL for cups and lids selected for examination in 4.2.2 shall be 4.0 percent defective. AQL for cups and lids selected for testing in 4.2.3 shall be 2.5 expressed in defects per hundred units.

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4.3 Inspection.

4.3.1 Visual examination. Cups and lids selected in 4.2.2 shall be examined in accordance with table IV.

4.3.2 Dimensional examination. Cups and lids examined in 4.3.1 shall be measured dimensionally for compliance with tables I, II, or IIa, as applicable.

TABLE IV. Classification of defects, end item

Examine	Defects
Cups:	
Design and construction	Item, type, and size not as specified. Drinking edge not rolled or rounded. Stacking feature missing. Visibly and detrimentally out of round. Not capable of being nested. Rounded bottoms; do not rest firmly on flat surface.
Workmanship	Not clean; any disagreeable odor. Any warpage, cracks, chips, tears, holes, butts, blisters, stains, discolorations, and flash.
Lids:	
Design and construction	Not coverall type. Do not fit cups snugly. Visibly and detrimentally out of round.
Workmanship	Not clean; any disagreeable odor. Any hole, tear, cut, wrinkle, crease, or fold. Rough edges.
Cups and lids:	
Identification	Marking not according to manufacturer's standard practice.

4.3.3 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packing, packaging, and marking requirements comply with section 5. Defects shall be scored in accordance with table V. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

TABLE V. Examination of preparation for delivery

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.

4.4 Tests.

4.4.1 Retest at standard atmospheric conditions (extra insulated cups). If failure is observed in conducting tests as specified in 4.4.2 through 4.4.6, inclusive, without standard atmospheric conditioning for item 1, type II cups, a retest shall be performed in accordance with ASTM E 171.

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4.4.2 Insulation (extra insulated cups). Fill cups with a liquid having a pour temperature of 180°F. ± 2°F. Measure the outside temperature of the cups with a thermocouple for five minutes after filling the cups. The cups shall not exceed the maximum allowable outside temperature requirements of 3.4.

4.4.3 Rigidity (extra insulated cups). Measure the outer brim cup diameter. Place the cup on its side and apply a vertical 200 gram force with a spherical end applicator one inch below the brim on the cup element opposite the side resting on a horizontal flat surface. Measure the deformed outer-brim cup diameter. The outer brim deflection shall not exceed the requirement of 3.5.

4.4.4 Staining tests, cups.

4.4.4.1 Brewed coffee. Fill the cups with brewed coffee having a pour temperature of 180°F. ± 2°F. Allow contents to remain in cup for 30 minutes. Pour out contents and rinse cup in cool, clear water. The sidewall shall meet the requirements of 3.6.1.

4.4.4.2 Carbonated cola beverage. Place several ice cubes in cup. Fill cup with a carbonated cola beverage and allow contents to remain in cup for one hour. Pour out contents and rinse cup in cool, clear water. The sidewall shall meet the requirement of 3.6.2.

4.4.5 Penetration, cups. Examine outer surfaces of cups tested in 4.4.4.1 and 4.4.4.2 just prior to pouring out contents. The outer surfaces shall meet the requirements of 3.7.

4.4.6 Distortion, cups. When pouring out the contents as specified in 4.4.4.1, simulate a drinking operation by lifting the cup approximately one foot with normal applied hand pressure. Hold the cup for one minute and empty the contents. The cup shall meet the requirement of 3.8.

4.4.7 Odor, cups and lids. Smell sample cups and lids at time of unpacking from boxes or bags (see 3.9).

4.4.8 Capacity, cups. Determine the cup capacity by measuring the volume of tap water, at room temperature, required to fill the cup to the brim (brimful). The minimum capacity of cups shall conform to the requirements of 3.10.

4.4.9 Thickness, cups and lids. Determine minimum thickness of cups and lids by using an instrument capable of measuring in mils and suitable for plastic material. Cups shall conform to the applicable requirements of tables I, II and IIa. Lids shall meet the thickness requirement of 3.3.2.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cups. Cups of one type and size in quantities specified in table VI, as applicable, shall be nested together and packaged in a close-fitting box conforming to PPP-B-566 or PPP-B-676. Alternatively, the cups may be packaged in a polyethylene bag and the bag adequately closed to prevent accidental opening.

5.1.1.2 Lids. The lids, in quantities specified in table VI, shall be packaged in a close-fitting box conforming to PPP-B-566, PPP-B-665, or PPP-B-676. Alternatively, the lids may be packaged in a polyethylene bag and the bag adequately closed to prevent accidental opening.

5.1.2 Level B. The cups and lids shall be packaged as specified in 5.1.1.1 and 5.1.1.2, respectively.

5.1.3 Level C. The cups and lids shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

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5.2.1.1 Cups. Cups of one type and size, in quantities specified in table VI, as applicable, shall be packed in a close-fitting box conforming to PPP-B-636, V3c. The box shall be closed and waterproofed in accordance with the appendix to the box specification.

5.2.1.2 Lids. The lids, in quantities specified in table VI, shall be packed in a close-fitting box conforming to PPP-B-636, V3c. The box shall be closed and waterproofed in accordance with the appendix to the box specification.

5.2.2 Level B. The cups and lids shall be packed as specified in 5.2.1.1 and 5.2.1.2, except that waterproof sealing shall not be required and the boxes shall conform to PPP-B-636, class domestic. Closure of boxes shall be in accordance with method II of the appendix to PPP-B-636.

5.2.3 Level C. The cups and lids shall be packed to insure protection during shipment and safe delivery to destination in compliance with the National Motor Freight Classification rules or the Uniform Freight Classification rules, as applicable.

5.3 Marking. In addition to any special markings required by the contract or order, all marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.2).

5.3.1 Civil agencies. All markings shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. All markings shall be in accordance with MIL-STD-129.

TABLE VI. Packaging and packing

Nominal size	Cups			Lids		
	Item and type	Packaging	Packing	Item	Packaging	Packing
Fluid oz.		Unit box or bag	Shipping carton		Unit box or bag	Shipping carton
6	1, II	25/50	1000	2	50/100	1000
7	1, I	75	1500	2	75	1500
8	1, II	25/50	1000	2	50/100	1000
10	1, II	25/50	1000	2	50/100	1000

6. NOTES

6.1 Intended use. Cups and lids covered by this specification are intended for single service and suitable for human consumption of hot and cold liquids.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Item, type, size, of cups (see 1.2).
(c) Size of cups for which lids are intended (see 1.2).
(d) Size of lot, if different from 4.2.1.
(e) Level of packaging and packing required (5.1 and 5.2).
(f) Marking required (5.3).

6.3 In order to insure proper fit of lids, it is necessary that they be ordered from the manufacturer of cups for which they are intended.

Preparing activity:

GSA-FSS

Civil Agency Coordinating Activities:

GSA-FSS
HEW-NIH
VA-DMS

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